Multiple Imputation Strategy for Time Alive and Out of Hospital (TAOOH) Outcome

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Overview

This document describes the multiple imputation strategy for the Time Alive and Out of Hospital (TAOOH) outcome in the LIQPLAT trial. The approach uses Multiple Imputation by Chained Equations (MICE) with multilevel predictive mean matching (PMM) to handle missing data under the Missing At Random (MAR) assumption.

The TAOOH outcome is a 5-level ordinal score constructed from the imputed binary indicators: (1) Alive and Out of Hospital, (2) Alive and In Hospital (Planned Admission), (3) Alive and In Hospital (ER Visit), (4) Alive and In Hospital (Unplanned Admission), (5) Dead. This ordinal score is derived *after* imputation of the component indicators.

Setup

```r
library(here)
library(tidyverse)
library(mice)
library(miceadds)
library(arrow)
library(data.table)
```

Data

The following code uses historical data from the Division of Medical Oncology at USB (first appointments from 2023 to mid-2024, prior to LIQPLAT initiation). These data are used to develop and validate the imputation procedure; actual trial data are not publicly available.

```r
set.seed(1234)
df_raw <- read_parquet(here("data", "long_data.parquet"))
```

Data Preprocessing

Variable Selection and Transformations

```r
# Create random treatment allocation for historical data
tx <- sample(unique(df_raw$pat_id), size = 0.5 * length(unique(df_raw$pat_id)), replace
= FALSE)

df_raw <- df_raw |>
  mutate(tx = if_else(pat_id %in% tx, "1", "0"))

imputation_vars <- c(
  "pat_id",
  "tx",
  "gender",
  "ecog_fstcnt",
  "diagnosis",
  "age",
  "albumin",
  "bilirubin",
  "c_reactive_protein",
  "lactate_dehydrogenase",
  "day",
  "in_hospital",
  "unplanned_stay",
  "er_visit",
  "death"
)

df <- df_raw |>
  select(all_of(imputation_vars)) |>
  mutate(
    diagnosis = fct_lump_n(diagnosis, n = 4),
    diagnosis = factor(as.integer(diagnosis)),
    ecog_fstcnt = factor(ecog_fstcnt, levels = 1:3, ordered = TRUE), #In LIQPLAT there
might be patients with level 4
    in_hospital = case_when(
        in_hospital == 1 ~ TRUE,
        TRUE ~ FALSE),
    unplanned_stay = case_when(
        unplanned_stay == 1 ~ TRUE,
        TRUE ~ FALSE),
    er_visit = case_when(
        er_visit == 1 ~ TRUE,
        TRUE ~ FALSE),
    lactate_dehydrogenase = log(lactate_dehydrogenase + 1),
```

```r
    tx = factor(tx)
    ) |>
  filter(day < 183) |>
  group_by(pat_id) |>
  mutate(
    survival_time = max(day),
    died = if_else(any(death == 1), 1, 0),
    week = ceiling(day / 7),
    survival_time = ceiling(survival_time / 7)
  ) |>
  group_by(pat_id, week) |>
  mutate(
    death = factor(max(death)),
    in_hospital = factor(max(in_hospital)),
    unplanned_stay = factor(max(unplanned_stay)),
    er_visit = factor(max(er_visit))
  ) |>
  ungroup() |>
  filter(week > -2) |>
  select(-day) |>
  distinct() |>
  arrange(pat_id, week)

# Model non-linear time effects using restricted cubic splines with 4 knots
basis_rcs <- rms::rcs(df$week, 4)

df$week <- as.vector(basis_rcs[, 1])
df$week_nl1 <- as.vector(basis_rcs[, 2])
df$week_nl2 <- as.vector(basis_rcs[, 3])
```

Nelson-Aalen Cumulative Hazard Estimate

We include the Nelson-Aalen cumulative hazard estimate (`na_est`) as a predictor for baseline variable imputation rather than using the `death` indicator directly. When imputing baseline variables using `2lonly.pmm`, longitudinal predictors are aggregated to patient-level means. We opted for the Nelson-Aalen cumulative hazard estimate:

[justification] 1. Is the recommended approach for including survival information in imputation models [vanBuuren2021-pc, chapter 9.2]

```r
na_df <- df |>
    select(pat_id, survival_time, died) |>
    unique()

na_est <- cbind(na_df$pat_id, nelsonaalen(na_df, timevar = survival_time, statusvar =
died)) |>
    data.table() |>
    rename(pat_id = V1, na_est = V2)

df <- df |>
    left_join(na_est, by = "pat_id")
```

Simulating Missing Data for Validation

For validation purposes, we introduce 10% missing data in the longitudinal outcome variables. In actual trial data, missingness in these variables is expected to be minimal.

```
df <- df |>
  mutate(
    is_missing = rbinom(n = n(), size = 1, prob = 0.1)
  ) |>
  mutate(
    er_visit = if_else(is_missing == 1, NA, er_visit),
    unplanned_stay = if_else(is_missing == 1, NA, unplanned_stay),
    in_hospital = if_else(is_missing == 1, NA, in_hospital)
  ) |>
    select(-is_missing)
```

Imputation Variables

Following Mathur2025-sl, we restrict the imputation model to variables included in the analysis model or those acting causally on the incomplete variables or their missingness indicators.

Analysis Variables

Variable	Description	Level
`tx`	Treatment group allocation	Baseline
`ecog_fstcnt`	ECOG performance status at baseline (0–4)	Baseline
`diagnosis`	Primary cancer diagnosis (5 categories)	Baseline
`albumin`	Serum albumin at baseline (g/L)	Baseline
`c_reactive_protein`	C-reactive protein at baseline (mg/L)	Baseline
`week`	Week after randomisation (RCS with 4 knots)	Longitudinal
`in_hospital`	Indicator for hospitalisation	Longitudinal
`unplanned_stay`	Indicator for unplanned hospital admission	Longitudinal
`er_visit`	Indicator for emergency room visit	Longitudinal
`death`	Death indicator	Longitudinal
`na_est`	Nelson-Aalen cumulative hazard estimate	Baseline

Table 1: Analysis variables included in the imputation model

Auxiliary Variables

Variable	Description	Level
`age`	Age at date of randomisation (years)	Baseline
`gender`	Patient sex	Baseline
`bilirubin`	Serum bilirubin at baseline (μmol/L)	Baseline
`lactate_dehydrogenase`	Lactate dehydrogenase at baseline (U/L, log-transformed)	Baseline

Table 2: Auxiliary variables included in the imputation model

Multiple Imputation Procedure

We employ multilevel multiple imputation to account for the hierarchical data structure, where repeated weekly observations (Level 1) are nested within patients (Level 2). The patient identifier (`pat_id`) serves as the clustering variable.

Imputation Methods

Baseline variables (Level 2) are imputed using `2lonly.pmm` from the `miceadds` package. This method:

1. Aggregates longitudinal predictors to the cluster level by computing patient means
2. Fits a linear regression model on the collapsed data:

$$Y_j = \beta_0 + \beta_1 \mathbf{X}_j + \beta_2 \bar{\mathbf{Z}}_j + \epsilon_j$$

   where Y_j is the missing baseline variable for patient j, $\mathbf{X}_j$ represents other baseline predictors, and $\bar{\mathbf{Z}}_j$ represents aggregated means of longitudinal predictors
3. Selects imputed values via predictive mean matching

Longitudinal variables (Level 1) are imputed using `2l.pmm`, which fits a linear mixed-effects model:

$$Y_{ij} = \beta_0 + \beta_1 \mathbf{X}_{ij} + u_{0j} + \epsilon_{ij}$$

where Y_{ij} is the outcome for observation i of patient j, $\mathbf{X}_{ij}$ includes fixed effects (time-varying and static predictors), and $u_{0j} \sim N(0, \sigma_u^2)$ is the patient-specific random intercept.

For both methods, we use Type 1 predictive mean matching with a donor pool of $k = 5$ [van-Buuren2021-pc, chapters 3.4, 7.3, 7.8].

Predictor Matrix Configuration

The predictor matrix is configured to:

- Use `pat_id` as the clustering variable (coded as -2)
- Exclude death-related variables from predictions (we use `na_est` instead)
- Prevent time variables from predicting static baseline variables

```r
md.pattern(df)
```

```r
init <- mice(df, maxit = 0)
pred_matrix <- init$predictorMatrix
meth <- init$method

# Use patient ID as clustering variable
pred_matrix[, "pat_id"] <- -2
pred_matrix["pat_id", ] <- 0

# Exclude raw death/survival variables from predictions (we use na_est instead)
# This avoids aggregation artefacts when collapsing longitudinal data to cluster means
pred_matrix[, "death"] <- 0
pred_matrix["death", ] <- 0
pred_matrix[, "died"] <- 0
pred_matrix["died", ] <- 0
pred_matrix[, "survival_time"] <- 0
pred_matrix["survival_time", ] <- 0

# week shouldn't predict static baseline variables nor itself
```

```r
baseline_vars <- c("tx", "age", "gender", "diagnosis", "ecog_fstcnt", "albumin",
"bilirubin",
                   "c_reactive_protein", "lactate_dehydrogenase", "na_est")

pred_matrix[c(baseline_vars, "week", "week_nl1", "week_nl2"), c("week", "week_nl1",
"week_nl2")] <- 0

# treatment, gender, age, and week are be missing
meth["tx"] <- ""
meth["gender"] <- ""
meth["age"] <- ""
meth["week"] <- ""
meth["week_nl1"] <- ""
meth["week_nl2"] <- ""


# use mean matching at the cluster level for baseline observations
# this ensures only one ecog per patient per imputation
meth["diagnosis"] <- "2lonly.pmm"
meth["ecog_fstcnt"] <- "2lonly.pmm"
meth["albumin"] <- "2lonly.pmm"
meth["bilirubin"] <- "2lonly.pmm"
meth["c_reactive_protein"] <- "2lonly.pmm"
meth["lactate_dehydrogenase"] <- "2lonly.pmm"

# Longitudinal variables: multilevel PMM
meth["in_hospital"] <- "2l.pmm"
meth["unplanned_stay"] <- "2l.pmm"
meth["er_visit"] <- "2l.pmm"


imputed_data <- mice(df,
                     m = 5,       # 50 in final analysis
                     maxit = 5,   # 50 in final analysis
                     predictorMatrix = pred_matrix,
                     method = meth,
                     donors = 5,
                     seed = 1234)
```

Imputation Parameters

For the final trial analysis:

- Number of imputed datasets: $m = 50$
- Maximum iterations: 50
- Donor pool size: $k = 5$

Convergence Diagnostics

Convergence is assessed via trace plots of the mean and standard deviation of imputed values across iterations.

```r
plot(imputed_data)
```

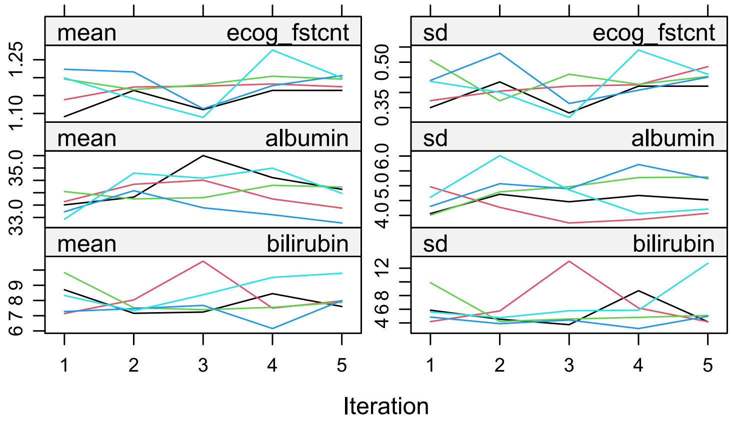


Figure 1: Trace plots showing the mean and standard deviation of imputed values across iterations for each imputed dataset.

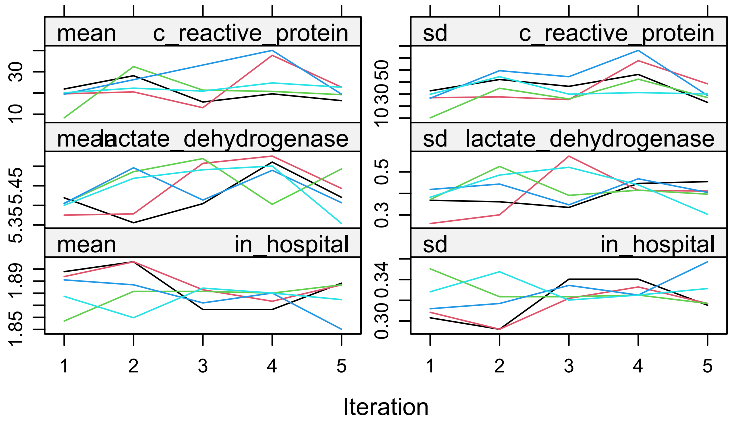


Figure 2: Trace plots showing the mean and standard deviation of imputed values across iterations for each imputed dataset.

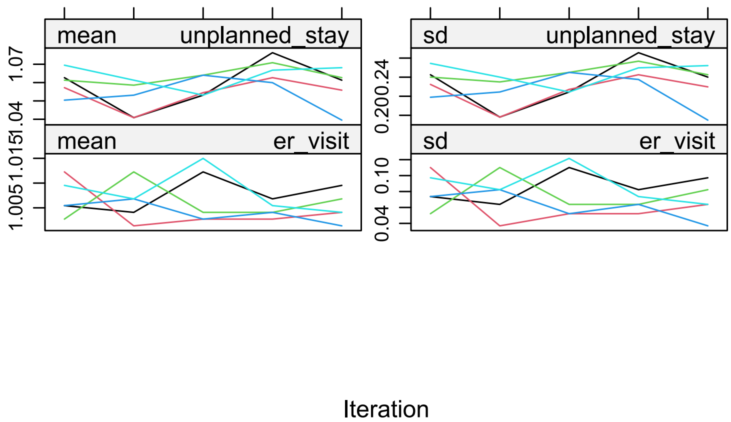


Figure 3: Trace plots showing the mean and standard deviation of imputed values across iterations for each imputed dataset.

Distribution Diagnostics

We compare the distributions of observed and imputed values to assess plausibility.

```
densityplot(imputed_data)
```

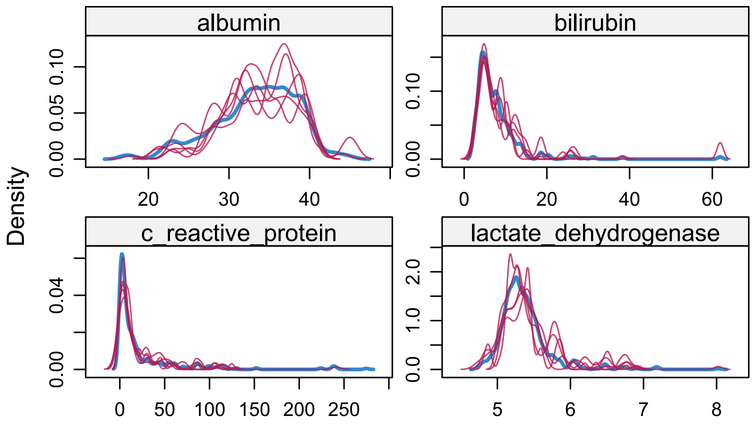


Figure 4: Density plots comparing observed (blue) and imputed (red) values for all imputed variables.

```
densityplot(imputed_data, ~ecog_fstcnt)
```

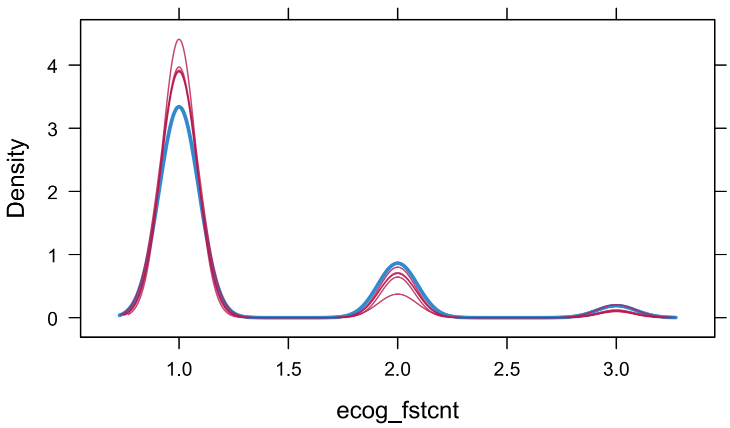
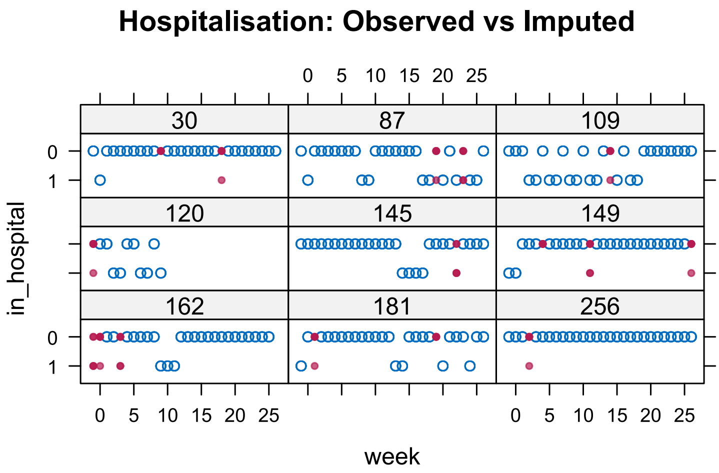


Figure 5: Density plot for ECOG performance status.

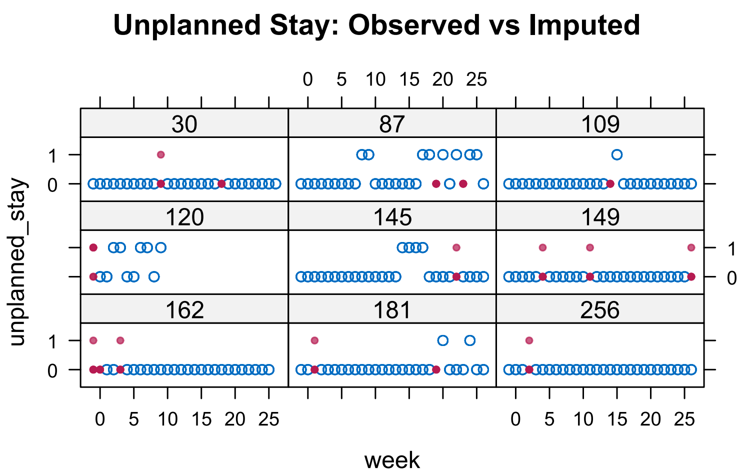
```
set.seed(1234)
original_data <- imputed_data$data
missing_pat_ids <- unique(original_data$pat_id[is.na(original_data$in_hospital)])
sample_ids <- sample(missing_pat_ids, 9)

xyplot(imputed_data,
       in_hospital ~ week | factor(pat_id),
       subset = pat_id %in% sample_ids,
       pch = c(1, 20),
       type = c("p"),
       lwd = 0,
```

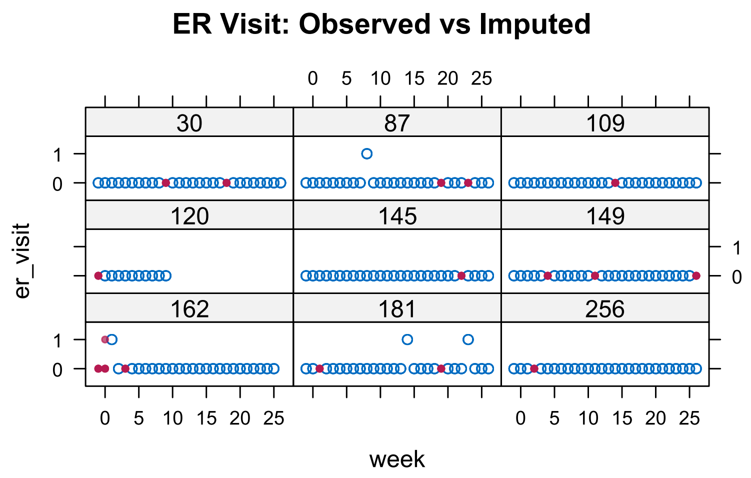
```
        layout = c(3, 3),
        main = "Hospitalisation: Observed vs Imputed")
```




```
xyplot(imputed_data,
       unplanned_stay ~ week | factor(pat_id),
       subset = pat_id %in% sample_ids,
       pch = c(1, 20),
       type = c("p"),
       lwd = 0,
       layout = c(3, 3),
       main = "Unplanned Stay: Observed vs Imputed")
```




```
xyplot(imputed_data,
       er_visit ~ week | factor(pat_id),
       subset = pat_id %in% sample_ids,
       pch = c(1, 20),
       type = c("p"),
       lwd = 0,
       layout = c(3, 3),
       main = "ER Visit: Observed vs Imputed")
```



Constructing the Ordinal TAOOH Score

After imputation of the component binary indicators, we construct a 5-level ordinal score representing the patient's weekly health state. The ordinal score is derived deterministically from the imputed indicators according to the following hierarchy:

Score	State	Condition
5	Dead	`death == 1`
4	Unplanned Hospitalisation	`in_hospital == 1` and `unplanned_stay == 1`
3	Emergency Room Visit	`er_visit == 1`
2	Planned Hospitalisation	`in_hospital == 1` and `unplanned_stay == 0`
1	Alive and Out of Hospital	Otherwise

Table 3: Definition of the 5-level ordinal TAOOH score

Analysis of Trial Data

For the final trial analysis, all $m = 50$ imputed datasets will be analysed, and posterior draws will be stacked to propagate imputation uncertainty into the final inference.

Simulation Study Dataset

For simulation studies, a single completed dataset is extracted.

```r
complete_data <- complete(imputed_data, 4)
```

Export

```r
write_parquet(complete_data, sink = here("data", "hosp_long_complete.parquet"))
```